Input-output equation from differential algebra approach

In[•]:=
```
(* Define equations*)
sys1 = {D[n1[x, t], t] + v1 * D[n1[x, t], x] +
     l1 * n1[x, t] - l2 * p21 * n2[x, t] - l3 * p31 * n3[x, t],
        D[n2[x, t], t] + v2 * D[n2[x, t], x] -
     l1 * p12 * n1[x, t] + l2 * n2[x, t] - l3 * (1 - p31) * n3[x, t],
        D[n3[x, t], t] + v3 * D[n3[x, t], x] -
     l1 * (1 - p12) * n1[x, t] - l2 * (1 - p21) * n2[x, t] + l3 * n3[x, t]};
```

In[•]:=
```
(* Obtain parameters in the above expression *)
params = Quiet[Complement[Variables[sys1], Join[{n1[x, t], n2[x, t], n3[x, t ]},
     Select[Variables[sys1], #[[0]][[1]] === n1 || #[[0]][[1]] === n2 || #[[0]][[1]] === n3 &]]]]
```

Out[•]=
```
{l1, l2, l3, p12, p21, p31, v1, v2, v3}
```

In[•]:=
```
(* Define observation function - LHS is the parameter to get rid of first *)
obs = n1[x, t] → n[x, t] - n2[x, t] - n3[x, t];
```

In[•]:=
```
sys2 = sys1 /. obs;
sys3 = sys2 /. D[obs, {x, 1}];
sys4 = FullSimplify[sys3 /. D[obs, {t, 1}]]
```

Out[•]=
$$\Big\{l1\, n[x,t] - (l1 + l2\, p21)\, n2[x,t] - (l1 + l3\, p31)\, n3[x,t] + n^{(0,1)}[x,t] - n2^{(0,1)}[x,t] - n3^{(0,1)}[x,t] + v1\left(n^{(1,0)}[x,t] - n2^{(1,0)}[x,t] - n3^{(1,0)}[x,t]\right),\\ -l1\, p12\, n[x,t] + (l2 + l1\, p12)\, n2[x,t] + (l1\, p12 + l3\,(-1 + p31))\, n3[x,t] + n2^{(0,1)}[x,t] + v2\, n2^{(1,0)}[x,t],\\ l2\,(-1 + p21)\, n2[x,t] + l1\,(-1 + p12)\,(n[x,t] - n2[x,t] - n3[x,t]) + l3\, n3[x,t] + n3^{(0,1)}[x,t] + v3\, n3^{(1,0)}[x,t]\Big\}$$

In[•]:=
```
summedsys = FullSimplify[sys1[[1]] + sys1[[2]] + sys1[[3]] /.
    Table[D[obs, {t, i[[1]]}], {x, i[[2]]}], {i, {{0, 0}, {1, 0}, {0, 1}}}]]
```

Out[•]=
$$n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] + (-v1 + v2)\, n2^{(1,0)}[x,t] + (-v1 + v3)\, n3^{(1,0)}[x,t]$$

In[•]:=
```
solsummedsys = Solve[0 == summedsys, D[n2[x, t], {x, 1}]][[1]]
```

Out[•]=
$$\left\{n2^{(1,0)}[x,t] \to \frac{n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t] + v3\, n3^{(1,0)}[x,t]}{v1 - v2}\right\}$$

In[•]:=
```
sys5 = D[sys4, {x, 1}] /. solsummedsys;
```

In[•]:=
```
sys6 = sys5 /. D[solsummedsys, {x, 1}];
```

In[•]:=
```
sys7 = Flatten[sys6 /. D[solsummedsys, {t, 1}]]
```

Out[•]=

$$\Big\{l1\, n^{(1,0)}[x,t] - (l1 + l3\, p31)\, n3^{(1,0)}[x,t] - \frac{(l1 + l2\, p21)\left(n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t] + v3\, n3^{(1,0)}[x,t]\right)}{v1 - v2} + n^{(1,1)}[x,t] - n3^{(1,1)}[x,t] - \frac{n^{(0,2)}[x,t] + v1\, n^{(1,1)}[x,t] - v1\, n3^{(1,1)}[x,t] + v3\, n3^{(1,1)}[x,t]}{v1 - v2} + v1\left(n^{(2,0)}[x,t] - n3^{(2,0)}[x,t] - \frac{n^{(1,1)}[x,t] + v1\, n^{(2,0)}[x,t] - v1\, n3^{(2,0)}[x,t] + v3\, n3^{(2,0)}[x,t]}{v1 - v2}\right),$$

$$-l1\, p12\, n^{(1,0)}[x,t] + (l1\, p12 + l3\,(-1 + p31))\, n3^{(1,0)}[x,t] + \frac{(l2 + l1\, p12)\left(n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t] + v3\, n3^{(1,0)}[x,t]\right)}{v1 - v2} + \frac{n^{(0,2)}[x,t] + v1\, n^{(1,1)}[x,t] - v1\, n3^{(1,1)}[x,t] + v3\, n3^{(1,1)}[x,t]}{v1 - v2} + \frac{v2\left(n^{(1,1)}[x,t] + v1\, n^{(2,0)}[x,t] - v1\, n3^{(2,0)}[x,t] + v3\, n3^{(2,0)}[x,t]\right)}{v1 - v2},$$

$$l3\, n3^{(1,0)}[x,t] + \frac{l2\,(-1 + p21)\left(n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t] + v3\, n3^{(1,0)}[x,t]\right)}{v1 - v2} + l1\,(-1 + p12)\left(n^{(1,0)}[x,t] - n3^{(1,0)}[x,t] - \frac{n^{(0,1)}[x,t] + v1\, n^{(1,0)}[x,t] - v1\, n3^{(1,0)}[x,t] + v3\, n3^{(1,0)}[x,t]}{v1 - v2}\right) + n3^{(1,1)}[x,t] + v3\, n3^{(2,0)}[x,t]\Big\}$$

In[•]:=
```
(* Define required order - the maximum needed should be 8 *)
requiredOrder = 4;
```

In[•]:=
```
(* The other variable(s) to eliminate *)
varElim = {n3}; varObs = n;
```

In[•]:=
```
(* All possible orderDifference derivatives to take *)
deriv = Flatten[Table[{i, j}, {i, 0, requiredOrder}, {j, 0, requiredOrder}], 1];
derivFull = Select[deriv, Total[#] ≤ requiredOrder &];
derivHigh = Select[deriv, Total[#] ≤ requiredOrder - 2 &];(* -1-1*)
derivLow = Select[deriv, Total[#] ≤ requiredOrder - 3 &];(* -2-1*)
```

In[•]:=
```
(* Unknown variables up to the required order *)
varUnknown = Flatten[
   Table[D[var[x, t], {x, d[[1]]}, {t, d[[2]]}], {d, derivFull}, {var, varElim}]];
```

In[•]:=
```
(* Expand each equation - only 2 and 3 cos the first one is -2-3 *)
sys8 = Flatten[Join[Table[D[sys7[[2]], {x, d[[1]]}, {t, d[[2]]}], {d, derivLow}],
    Table[D[sys7[[3]], {x, d[[1]]}, {t, d[[2]]}], {d, derivLow}]]];
```

In[•]:=
```
(* Create matrix system *)
M = Table[Coefficient[expr, var], {expr, sys8}, {var, varUnknown}];
b = FullSimplify[M.varUnknown - sys8];
```

In[•]:=
```
(* Determine number of polynomial expressions required *)
Mb = Join[M, ArrayReshape[b, {Length[M], 1}], 2]
numExpr = MatrixRank[Mb] - MatrixRank[M]
```

Out[ ]=

$$\left\{\left\{0,\,0,\,0,\,0,\,0,\,-l3+l1\,p12+l3\,p31-\frac{l2\,v1}{v1-v2}-\frac{l1\,p12\,v1}{v1-v2}+\frac{l2\,v3}{v1-v2}+\frac{l1\,p12\,v3}{v1-v2},\right.\right.$$
$$-\frac{v1}{v1-v2}+\frac{v3}{v1-v2},\,0,\,0,\,-\frac{v1\,v2}{v1-v2}+\frac{v2\,v3}{v1-v2},\,0,\,0,\,0,$$
$$0,\,0,\,-\frac{1}{v1-v2}\left((l2+l1\,p12)\,n^{(0,1)}[x,t]+n^{(0,2)}[x,t]+\right.$$
$$\left.\left.(l2\,v1+l1\,p12\,v2)\,n^{(1,0)}[x,t]+(v1+v2)\,n^{(1,1)}[x,t]+v1\,v2\,n^{(2,0)}[x,t]\right)\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,0,\,-l3+l1\,p12+l3\,p31-\frac{l2\,v1}{v1-v2}-\frac{l1\,p12\,v1}{v1-v2}+\frac{l2\,v3}{v1-v2}+\frac{l1\,p12\,v3}{v1-v2},\right.$$
$$-\frac{v1}{v1-v2}+\frac{v3}{v1-v2},\,0,\,0,\,-\frac{v1\,v2}{v1-v2}+\frac{v2\,v3}{v1-v2},\,0,\,0,\,0,\,0,$$
$$-\frac{1}{v1-v2}\left((l2+l1\,p12)\,n^{(0,2)}[x,t]+n^{(0,3)}[x,t]+(l2\,v1+l1\,p12\,v2)\,n^{(1,1)}[x,t]+\right.$$
$$\left.\left.(v1+v2)\,n^{(1,2)}[x,t]+v1\,v2\,n^{(2,1)}[x,t]\right)\right\},\left\{0,\,0,\,0,\,0,\,0,\,0,\,0,\,0,\,0,\right.$$
$$-l3+l1\,p12+l3\,p31-\frac{l2\,v1}{v1-v2}-\frac{l1\,p12\,v1}{v1-v2}+\frac{l2\,v3}{v1-v2}+\frac{l1\,p12\,v3}{v1-v2},\,-\frac{v1}{v1-v2}+\frac{v3}{v1-v2},$$
$$0,\,-\frac{v1\,v2}{v1-v2}+\frac{v2\,v3}{v1-v2},\,0,\,0,\,-\frac{1}{v1-v2}\left((l2+l1\,p12)\,n^{(1,1)}[x,t]+n^{(1,2)}[x,t]+\right.$$
$$\left.\left.(l2\,v1+l1\,p12\,v2)\,n^{(2,0)}[x,t]+(v1+v2)\,n^{(2,1)}[x,t]+v1\,v2\,n^{(3,0)}[x,t]\right)\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,l1+l3-l1\,p12-\frac{l1\,v1}{v1-v2}+\frac{l2\,v1}{v1-v2}+\frac{l1\,p12\,v1}{v1-v2}-\frac{l2\,p21\,v1}{v1-v2}+\right.$$
$$\frac{l1\,v3}{v1-v2}-\frac{l2\,v3}{v1-v2}-\frac{l1\,p12\,v3}{v1-v2}+\frac{l2\,p21\,v3}{v1-v2},\,1,\,0,\,0,\,v3,$$
$$0,\,0,\,0,\,0,\,0,\,\frac{1}{v1-v2}\left((l2+l1\,(-1+p12)-l2\,p21)\,n^{(0,1)}[x,t]+\right.$$
$$\left.\left.(l2\,(v1-p21\,v1)+l1\,(-1+p12)\,v2)\,n^{(1,0)}[x,t]\right)\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,0,\,l1+l3-l1\,p12-\frac{l1\,v1}{v1-v2}+\frac{l2\,v1}{v1-v2}+\frac{l1\,p12\,v1}{v1-v2}-\right.$$
$$\frac{l2\,p21\,v1}{v1-v2}+\frac{l1\,v3}{v1-v2}-\frac{l2\,v3}{v1-v2}-\frac{l1\,p12\,v3}{v1-v2}+\frac{l2\,p21\,v3}{v1-v2},\,1,\,0,\,0,$$
$$v3,\,0,\,0,\,0,\,0,\,\frac{1}{v1-v2}\left((l2+l1\,(-1+p12)-l2\,p21)\,n^{(0,2)}[x,t]+\right.$$
$$\left.\left.(l2\,(v1-p21\,v1)+l1\,(-1+p12)\,v2)\,n^{(1,1)}[x,t]\right)\right\},$$
$$\left\{0,\,0,\,0,\,0,\,0,\,0,\,0,\,0,\,0,\,l1+l3-l1\,p12-\frac{l1\,v1}{v1-v2}+\frac{l2\,v1}{v1-v2}+\frac{l1\,p12\,v1}{v1-v2}-\right.$$
$$\frac{l2\,p21\,v1}{v1-v2}+\frac{l1\,v3}{v1-v2}-\frac{l2\,v3}{v1-v2}-\frac{l1\,p12\,v3}{v1-v2}+\frac{l2\,p21\,v3}{v1-v2},\,1,$$
$$0,\,v3,\,0,\,0,\,\frac{1}{v1-v2}\left((l2+l1\,(-1+p12)-l2\,p21)\,n^{(1,1)}[x,t]+\right.$$
$$\left.\left.\left.(l2\,(v1-p21\,v1)+l1\,(-1+p12)\,v2)\,n^{(2,0)}[x,t]\right)\right\}\right\}$$

Out[ ]=

1

In[•]:=
```
(* Perform row reduction, extract polynomial expression *)
{walltime, op} = AbsoluteTiming[
   RowReduce[Join[M, IdentityMatrix[Length[M]], 2]][[ ;; , -Length[M] ;;]];
walltime
expr = FullSimplify[op[[-numExpr ;;, ;;]].b]
```

Out[•]=
```
0.811972
```

Out[•]=
$$\left\{-\left(\left(\left(l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)+l2\,l3\,(p21+p31-p21\,p31)\right)\right.\right.$$
$$n^{(0,1)}[x,t]+(l1+l2+l3)\,n^{(0,2)}[x,t]+n^{(0,3)}[x,t]+l2\,l3\,p21\,v1\,n^{(1,0)}[x,t]+$$
$$l2\,l3\,p31\,v1\,n^{(1,0)}[x,t]-l2\,l3\,p21\,p31\,v1\,n^{(1,0)}[x,t]+l1\,l3\,v2\,n^{(1,0)}[x,t]-$$
$$l1\,l3\,p31\,v2\,n^{(1,0)}[x,t]+l1\,l3\,p12\,p31\,v2\,n^{(1,0)}[x,t]+l1\,l2\,v3\,n^{(1,0)}[x,t]-$$
$$l1\,l2\,p12\,p21\,v3\,n^{(1,0)}[x,t]+l2\,v1\,n^{(1,1)}[x,t]+l3\,v1\,n^{(1,1)}[x,t]+$$
$$l1\,v2\,n^{(1,1)}[x,t]+l3\,v2\,n^{(1,1)}[x,t]+l1\,v3\,n^{(1,1)}[x,t]+l2\,v3\,n^{(1,1)}[x,t]+$$
$$v1\,n^{(1,2)}[x,t]+v2\,n^{(1,2)}[x,t]+v3\,n^{(1,2)}[x,t]+l3\,v1\,v2\,n^{(2,0)}[x,t]+$$
$$l2\,v1\,v3\,n^{(2,0)}[x,t]+l1\,v2\,v3\,n^{(2,0)}[x,t]+v1\,v2\,n^{(2,1)}[x,t]+$$
$$\left.v1\,v3\,n^{(2,1)}[x,t]+v2\,v3\,n^{(2,1)}[x,t]+v1\,v2\,v3\,n^{(3,0)}[x,t]\right)\Big/$$
$$\left.\left.(l3\,(v1-v2)-l2\,(-1+p21)\,(v1-v3)+l1\,(-1+p12)\,(v2-v3))\right)\right\}$$

In[•]:=
```
expr2 =
 FullSimplify[expr / (1/(l3 (-v1 + v2) + l2 (-1 + p21) (v1 - v3) - l1 (-1 + p12) (v2 - v3)))]
```

Out[•]=
$$\left\{(l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)+l2\,l3\,(p21+p31-p21\,p31))\,n^{(0,1)}[x,t]+\right.$$
$$(l1+l2+l3)\,n^{(0,2)}[x,t]+n^{(0,3)}[x,t]+l2\,l3\,p21\,v1\,n^{(1,0)}[x,t]+$$
$$l2\,l3\,p31\,v1\,n^{(1,0)}[x,t]-l2\,l3\,p21\,p31\,v1\,n^{(1,0)}[x,t]+l1\,l3\,v2\,n^{(1,0)}[x,t]-$$
$$l1\,l3\,p31\,v2\,n^{(1,0)}[x,t]+l1\,l3\,p12\,p31\,v2\,n^{(1,0)}[x,t]+l1\,l2\,v3\,n^{(1,0)}[x,t]-$$
$$l1\,l2\,p12\,p21\,v3\,n^{(1,0)}[x,t]+l2\,v1\,n^{(1,1)}[x,t]+l3\,v1\,n^{(1,1)}[x,t]+$$
$$l1\,v2\,n^{(1,1)}[x,t]+l3\,v2\,n^{(1,1)}[x,t]+l1\,v3\,n^{(1,1)}[x,t]+l2\,v3\,n^{(1,1)}[x,t]+$$
$$v1\,n^{(1,2)}[x,t]+v2\,n^{(1,2)}[x,t]+v3\,n^{(1,2)}[x,t]+l3\,v1\,v2\,n^{(2,0)}[x,t]+$$
$$l2\,v1\,v3\,n^{(2,0)}[x,t]+l1\,v2\,v3\,n^{(2,0)}[x,t]+v1\,v2\,n^{(2,1)}[x,t]+$$
$$\left.v1\,v3\,n^{(2,1)}[x,t]+v2\,v3\,n^{(2,1)}[x,t]+v1\,v2\,v3\,n^{(3,0)}[x,t]\right\}$$

In[•]:=
```
(* Extract coefficients *)
varKnown = Join[{n[x, t]}, Quiet[Select[Variables[expr2], #[[0]][[1]] === n &]]]
coef1 = FullSimplify[Flatten[Values@CoefficientRules[expr2, varKnown]]]
```

Out[•]=
$$\left\{n[x,t],\,n^{(0,1)}[x,t],\,n^{(0,2)}[x,t],\,n^{(0,3)}[x,t],\,n^{(1,0)}[x,t],\right.$$
$$\left.n^{(1,1)}[x,t],\,n^{(1,2)}[x,t],\,n^{(2,0)}[x,t],\,n^{(2,1)}[x,t],\,n^{(3,0)}[x,t]\right\}$$

Out[•]=
$$\{l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)+l2\,l3\,(p21+p31-p21\,p31),\,l1+l2+l3,\,1,$$
$$l1\,l3\,(1+(-1+p12)\,p31)\,v2+l2\,(l3\,(p21+p31-p21\,p31)\,v1+l1\,(1-p12\,p21)\,v3),$$
$$l3\,(v1+v2)+l2\,(v1+v3)+l1\,(v2+v3),\,v1+v2+v3,$$
$$l3\,v1\,v2+l2\,v1\,v3+l1\,v2\,v3,\,v2\,v3+v1\,(v2+v3),\,v1\,v2\,v3\}$$

```
(* Try another way: upper triangularise the matrix *)
{walltime2, op2} =
  AbsoluteTiming[LinearAlgebra`Private`UpperTriangularizeMatrix[
     Join[M, IdentityMatrix[Length[M]], 2]][[ ;; , -Length[M] ;;]]];
walltime2
```

Out[ ]=

```
0.154415
```

In[ ]:=

```
(* Compute expression *)
expr2 = FullSimplify[op2[[-numExpr ;;, ;;]].b]
```

Out[ ]=

$$\left\{\frac{1}{(v1-v2)^7}\,(l3\,(-1+p31)\,(v1-v2)+l2\,(-v1+v3)+l1\,p12\,(-v2+v3))^2\right.$$
$$(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+$$
$$l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2$$
$$\Big((l1\,(l2+l3-l2\,p12\,p21+l3\,(-1+p12)\,p31)+l2\,l3\,(p21+p31-p21\,p31))$$
$$n^{(0,1)}[x,t]+(l1+l2+l3)\,n^{(0,2)}[x,t]+n^{(0,3)}[x,t]+l2\,l3\,p21\,v1\,n^{(1,0)}[x,t]+$$
$$l2\,l3\,p31\,v1\,n^{(1,0)}[x,t]-l2\,l3\,p21\,p31\,v1\,n^{(1,0)}[x,t]+l1\,l3\,v2\,n^{(1,0)}[x,t]-$$
$$l1\,l3\,p31\,v2\,n^{(1,0)}[x,t]+l1\,l3\,p12\,p31\,v2\,n^{(1,0)}[x,t]+l1\,l2\,v3\,n^{(1,0)}[x,t]-$$
$$l1\,l2\,p12\,p21\,v3\,n^{(1,0)}[x,t]+l2\,v1\,n^{(1,1)}[x,t]+l3\,v1\,n^{(1,1)}[x,t]+$$
$$l1\,v2\,n^{(1,1)}[x,t]+l3\,v2\,n^{(1,1)}[x,t]+l1\,v3\,n^{(1,1)}[x,t]+l2\,v3\,n^{(1,1)}[x,t]+$$
$$v1\,n^{(1,2)}[x,t]+v2\,n^{(1,2)}[x,t]+v3\,n^{(1,2)}[x,t]+l3\,v1\,v2\,n^{(2,0)}[x,t]+$$
$$l2\,v1\,v3\,n^{(2,0)}[x,t]+l1\,v2\,v3\,n^{(2,0)}[x,t]+v1\,v2\,n^{(2,1)}[x,t]+$$
$$\left.v1\,v3\,n^{(2,1)}[x,t]+v2\,v3\,n^{(2,1)}[x,t]+v1\,v2\,v3\,n^{(3,0)}[x,t]\Big)\right\}$$

In[•]:=
```
(* Extract coefficients *)
coef12 = FullSimplify[Flatten[Values@CoefficientRules[expr2, varKnown]]]
```

Out[•]=

$$\left\{-\frac{1}{(v1-v2)^7}(l1\,l2\,(-1+p12\,p21)+l2\,l3\,p21\,(-1+p31)-l2\,l3\,p31+l1\,l3\,(-1+p31-p12\,p31))\right.$$
$$(-l3\,(-1+p31)\,(v1-v2)+l2\,(v1-v3)+l1\,p12\,(v2-v3))^2$$
$$(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+$$
$$l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2,$$
$$\frac{1}{(v1-v2)^7}(l1+l2+l3)\,(-l3\,(-1+p31)\,(v1-v2)+l2\,(v1-v3)+l1\,p12\,(v2-v3))^2$$
$$(-l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)+l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+$$
$$l1\,(v2-v3)\,(v1-p12\,v2+(-1+p12)\,v3))^2,$$
$$\frac{1}{(v1-v2)^7}(l3\,(-1+p31)\,(v1-v2)+l2\,(-v1+v3)+l1\,p12\,(-v2+v3))^2$$
$$(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-$$
$$l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2,$$
$$-\frac{1}{(v1-v2)^7}(l2\,l3\,(p21\,(-1+p31)-p31)\,v1+l1\,l3\,(-1+p31-p12\,p31)\,v2+l1\,l2$$
$$(-1+p12\,p21)\,v3)\,(l3\,(-1+p31)\,(v1-v2)+l2\,(-v1+v3)+l1\,p12\,(-v2+v3))^2$$
$$(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+$$
$$l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2,$$
$$\frac{1}{(v1-v2)^7}(l3\,(-1+p31)\,(v1-v2)+l2\,(-v1+v3)+l1\,p12\,(-v2+v3))^2$$
$$(l3\,(v1+v2)+l2\,(v1+v3)+l1\,(v2+v3))$$
$$(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+$$
$$l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2,$$
$$\frac{1}{(v1-v2)^7}(v1+v2+v3)\,(l3\,(-1+p31)\,(v1-v2)+l2\,(-v1+v3)+l1\,p12\,(-v2+v3))^2$$
$$(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+$$
$$l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2,$$
$$\frac{1}{(v1-v2)^7}(l3\,v1\,v2+l2\,v1\,v3+l1\,v2\,v3)$$
$$(l3\,(-1+p31)\,(v1-v2)+l2\,(-v1+v3)+l1\,p12\,(-v2+v3))^2$$
$$(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+$$
$$l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2,$$
$$\frac{1}{(v1-v2)^7}(l3\,(-1+p31)\,(v1-v2)+l2\,(-v1+v3)+l1\,p12\,(-v2+v3))^2$$
$$(v2\,v3+v1\,(v2+v3))\,(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-$$
$$l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2,$$
$$\frac{1}{(v1-v2)^7}v1\,v2\,v3\,(l3\,(-1+p31)\,(v1-v2)+l2\,(-v1+v3)+l1\,p12\,(-v2+v3))^2$$
$$(l3\,(v1-v2)\,(p31\,v1+v2-p31\,v2-v3)-$$
$$\left.l2\,(v1-v3)\,(-v2+p21\,(v1-v3)+v3)+l1\,(v2-v3)\,(-v1+p12\,v2+v3-p12\,v3))^2\right\}$$

The structurally identifiable coefficients from above are:

In[•]:=
```
coef22 = FullSimplify[coef12 / coef12[[3]]]
```

Out[•]=
```
{l1 (l2 + l3 - l2 p12 p21 + l3 (-1 + p12) p31) + l2 l3 (p21 + p31 - p21 p31), l1 + l2 + l3, 1,
 l1 l3 (1 + (-1 + p12) p31) v2 + l2 (l3 (p21 + p31 - p21 p31) v1 + l1 (1 - p12 p21) v3),
 l3 (v1 + v2) + l2 (v1 + v3) + l1 (v2 + v3), v1 + v2 + v3,
 l3 v1 v2 + l2 v1 v3 + l1 v2 v3, v2 v3 + v1 (v2 + v3), v1 v2 v3}
```